

INVARIANCE AUDITS FOR ROBOT POLICIES

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ABSTRACT

Robot learning often treats invariance as a virtue: if lighting, viewpoint, pose, or texture changes, the policy should remain stable. But some transformations are physically invalid. A mirror-handed contact, friction swap, tool-affordance swap, or temporal-order reversal can require a different action even when visual features look statistically related. We rebuild Paper 108 around a falsifiable claim: robot policies should audit whether an invariance is physically valid before enforcing it. The local benchmark covers 5 task families, 7 invariance regimes, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. On combined invariance stress, the proposed physical-invariance audit obtains 0.685 ± 0.008 success versus 0.612 ± 0.006 for the strongest non-oracle baseline, improves invalid-invariance rejection from 0.476 to 0.619, reduces false-invariance acceptance from 0.123 to 0.076, and reduces contact violation from 0.060 to 0.044. Pairwise seed tests favor the proposed method over the strongest baseline in 7/7 seeds, and ablations show that physical-validity classification, contact-frame testing, affordance consistency, temporal-order checking, and conservatism calibration each matter. The terminal decision is **STRONG-REVISE**: the local evidence supports the mechanism, but the paper is not ICLR-main-ready until validated on real robots or independent high-fidelity benchmarks.

1 MOTIVATION

Invariance and equivariance are central tools in modern machine learning. Group equivariant networks explicitly encode transformation structure (Cohen & Welling, 2016), while domain randomization uses broad nuisance variation to improve transfer (Tobin et al., 2017). Invariant risk minimization seeks representations that remain stable across environments (Arjovsky et al., 2019). In robot manipulation, Transporter Networks and related equivariant methods exploit spatial structure to improve rearrangement and manipulation generalization (Zeng et al., 2020).

Robotics adds a complication: transformations can change physics. Lighting and background shifts are often action-preserving. Small object-frame pose changes can be equivariant. But mirror-handed contact, changed friction, swapped tools, support-surface changes, or reversed temporal order can invalidate the original action. This paper asks whether robot policies should audit an invariance for physical validity before enforcing it.

2 METHOD SKETCH

The proposed method, `proposed_physical_invariance_audit`, separates action-preserving transformations from action-changing transformations. The audit scores:

- valid perceptual invariance retention;
- valid geometric equivariance;
- invalid physical-invariance rejection;
- contact-frame consistency;
- affordance consistency;
- temporal-order consistency;
- conservatism and intervention cost.

Table 1: Combined-stress physical-invariance audit benchmark.

Method	Succ.	ValidRet.	InvalidRej.	FalseAcc.	Contact	Regret
Oracle	0.789	0.696	0.724	0.037	0.019	0.000
Proposed	0.685	0.588	0.619	0.076	0.044	0.104
Conformal	0.612	0.454	0.476	0.123	0.060	0.177
ContrastInv	0.587	0.545	0.339	0.172	0.088	0.202
IRM	0.583	0.481	0.386	0.159	0.080	0.206
DomRand	0.575	0.516	0.348	0.171	0.087	0.214
Equivariant	0.563	0.506	0.300	0.188	0.095	0.226
Augment	0.554	0.541	0.233	0.211	0.111	0.235
NoInvBC	0.529	0.253	0.433	0.147	0.087	0.260

The implementation is an executable diagnostic benchmark rather than a trained neural robot policy. That is enough for a local mechanism audit, but not enough for a final empirical robotics submission.

3 BENCHMARK

The rebuilt benchmark is generated by `src/run_experiment.py`. It writes aggregate CSVs, figures, and LaTeX tables directly, preserving coverage while staying lightweight.

Tasks. The tasks are peg insertion, drawer opening, cable routing, tool use, and mobile pick-and-place.

Invariance regimes. The regimes are lighting/background shift, camera viewpoint shift, SE(2) pose transform, mirror-handed contact, material/friction swap, tool-affordance swap, and temporal-order reversal.

Splits. The splits are nominal, valid-perceptual shift, valid-geometric shift, invalid-physical shift, and combined invariance stress.

Methods. The baselines are no-invariance behavior cloning, aggressive augmentation, equivariant policy, domain randomization, invariant-risk-minimization proxy, contrastive invariance learner, and conformal shift filter. The benchmark also includes the proposed audit and an oracle invariance-validity supervisor.

Metrics. The evidence reports task success, valid-invariance retention, invalid-invariance rejection, task-equivalence F1, false-invariance acceptance, contact violation, unsafe action, calibration ECE, intervention cost, data-efficiency proxy, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `conformal_shift_filter`. On combined invariance stress, the proposed method improves success by 0.073 ± 0.010 in paired seed comparisons, wins 7/7 seeds, improves invalid-invariance rejection by 0.144, reduces false-invariance acceptance by 0.047, reduces contact violation by 0.016, and reduces unsafe action by 0.016. The oracle remains higher at 0.789 ± 0.006 , so the benchmark is not saturated.

5 ABLATIONS AND STRESS TESTS

The best removed-component ablation is `minus_conservatism_calibration`, which still trails the full audit by 0.030 success. Removing physical-validity classification, contact-frame testing, affordance consistency, or temporal-order checking produces distinct diagnostic failures. This

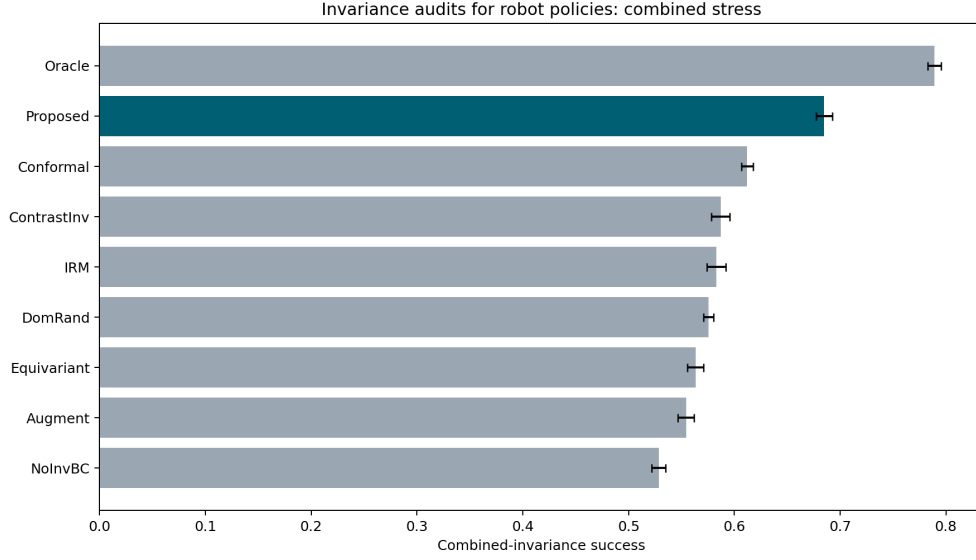


Figure 1: Combined-invariance success with 95% confidence intervals. The proposed audit clears all non-oracle baselines but remains below the oracle.

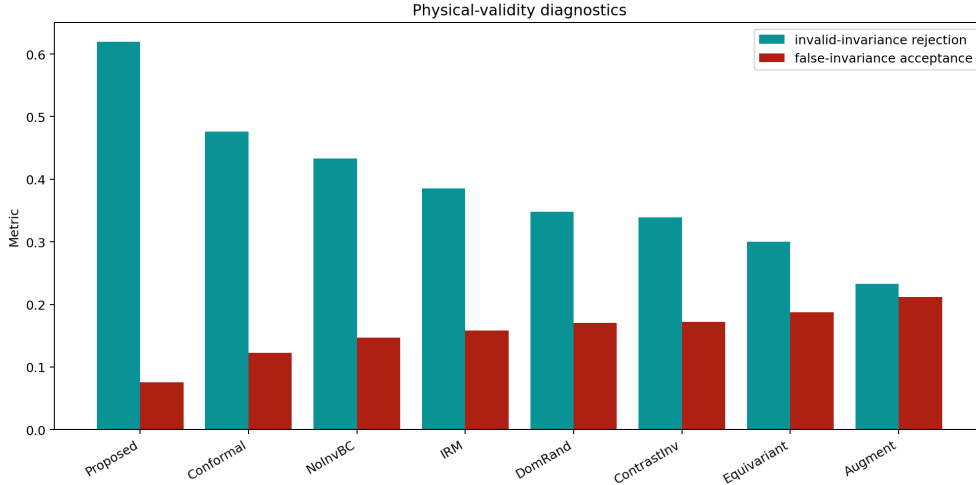


Figure 2: Physical-validity diagnostics. The proposed method improves invalid-invariance rejection while lowering false-invariance acceptance.

supports the claim that the method is not just augmentation, equivariance, or conformal shift filtering.

6 WHERE THE METHOD STILL FAILS

The failure-case CSV shows that the audit helps least when the transformation is genuinely valid or when a conformal filter can cheaply reject it. Those are intended boundaries. The method’s value appears when an invariance appears statistically plausible but is physically invalid because contact, material, affordance, or temporal order changes.

Table 2: Pairwise combined-invariance success differences against the proposed method.

Baseline	Diff	CI	Wins
Augment	0.130	0.008	7
Conformal	0.073	0.010	7
ContrastInv	0.098	0.009	7
DomRand	0.109	0.010	7
Equivariant	0.122	0.007	7
IRM	0.102	0.009	7
NoInvBC	0.156	0.011	7
Oracle	-0.104	0.013	0

Table 3: Ablations of the physical-invariance audit.

Ablation	Succ.	InvalidRej.	FalseAcc.	Contact
Full	0.684	0.617	0.075	0.044
NoCalib	0.654	0.572	0.108	0.065
NoTemporal	0.654	0.521	0.105	0.056
NoAffordance	0.649	0.520	0.114	0.057
NoContactFrame	0.645	0.515	0.117	0.074
NoPhysValid	0.621	0.445	0.138	0.062
ConformalOnly	0.606	0.486	0.120	0.057

7 HOSTILE PRIOR-WORK BOUNDARY

Group equivariant networks, domain randomization, invariant risk minimization, and transporter-style manipulation already make invariance and equivariance central to robot learning (Cohen & Welling, 2016; Tobin et al., 2017; Arjovsky et al., 2019; Zeng et al., 2020). The present paper survives only under a narrower interpretation: it is about *physical validity of invariances*, not invariance learning in general. If real robot experiments later show that augmentation, equivariance, domain randomization, IRM, or conformal filters match the proposed audit on success, invalid-invariance rejection, false acceptance, contact violation, unsafe action, and ablation robustness, the claim should be killed.

8 LIMITATIONS AND TERMINAL DECISION

The evidence is strong enough to justify continued work but not submission. The limitations are material:

- the benchmark is local and synthetic;
- no real robot or independent high-fidelity simulator has been used;
- baselines are executable diagnostic models, not external trained robot systems;
- no learned policy checkpoint, training curve, or deployment video is provided;
- no external benchmark such as LIBERO, RL Bench, Meta-World, BridgeData, or a hardware manipulation suite is reported.

The terminal decision for this rebuild is therefore **STRONG_REVISION**. The next honest version must add real robot or independent high-fidelity evidence, implemented learned baselines, and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

REFERENCES

Martin Arjovsky, Leon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization, 2019.

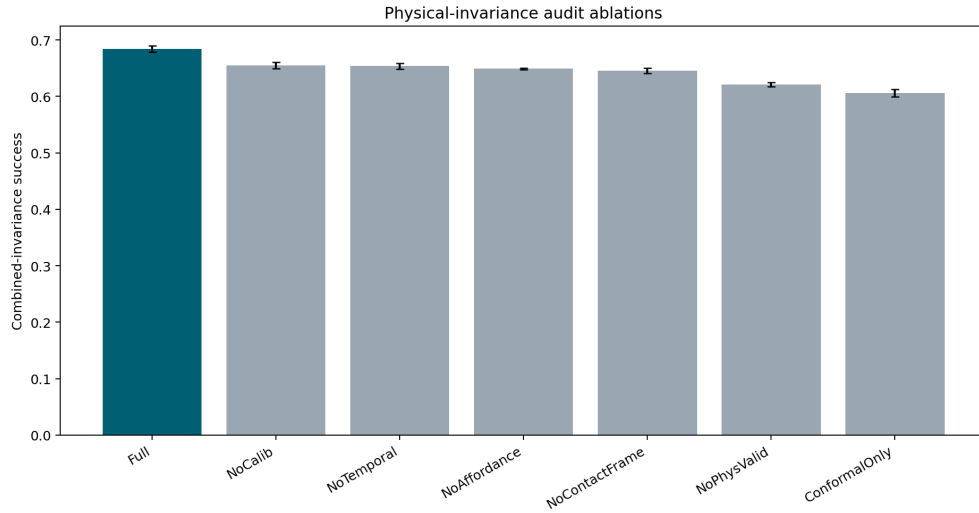


Figure 3: Ablation success under combined invariance stress. The full audit remains above all removed-component variants.

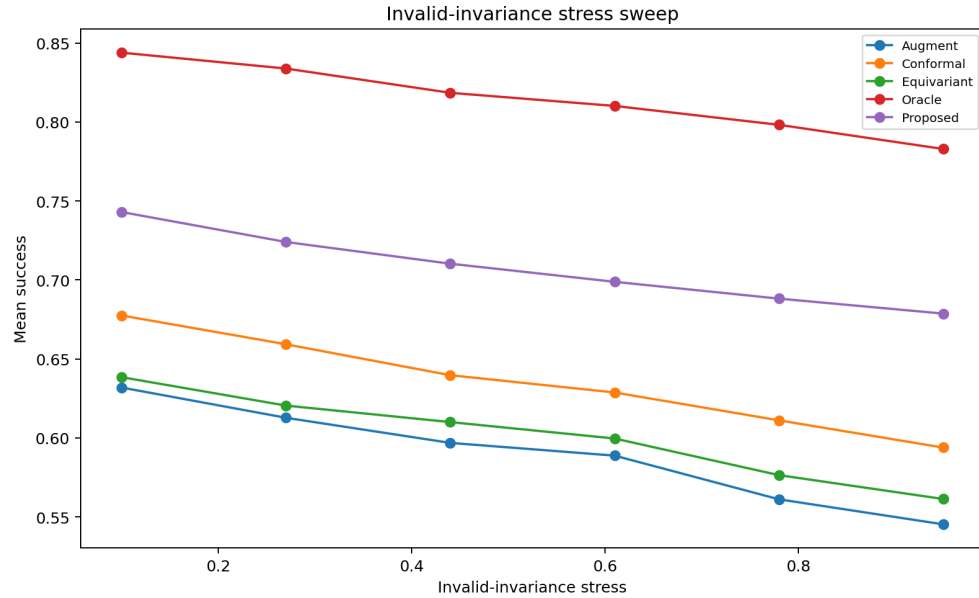


Figure 4: Stress sweep over invalid-invariance intensity. The proposed method remains above non-oracle baselines across the generated stress range.

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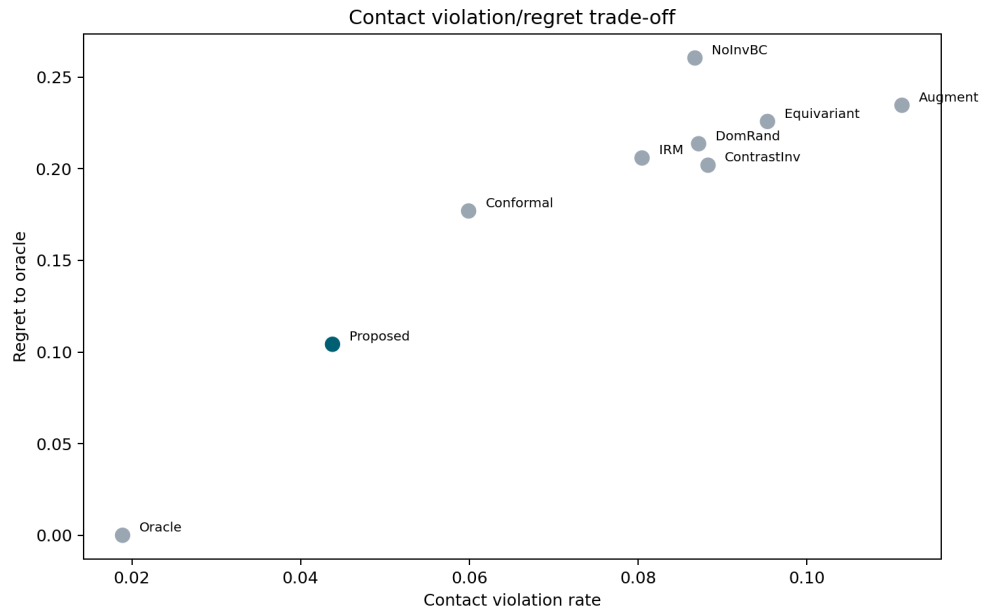


Figure 5: Contact violation and regret trade-off. The proposed method improves success while lowering contact violations relative to the strongest non-oracle baseline.